RF-Based Drone Identification Using Machine Learning

[23ET7PE3ML]

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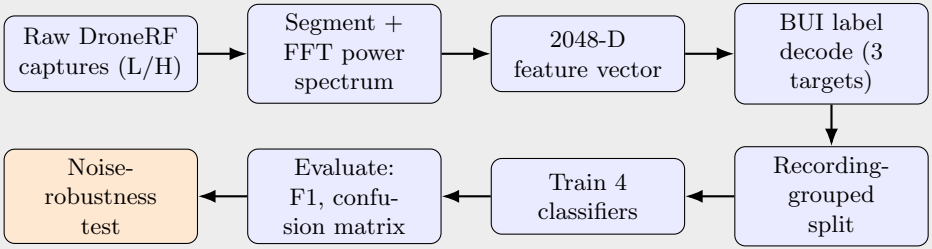
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Introduction and Problem Definition

- Drones communicate with their controllers over radio-frequency (RF) links in the 2.4 GHz band. Listening to these signals enables passive detection without line of sight, unlike cameras or radar.
- Detecting presence alone is insufficient; a practical system must also identify the drone type and, where possible, its operating condition.
- The 2.4 GHz band is shared with Wi-Fi and Bluetooth, so recorded signals are noisy and classification is non-trivial.
- Segments extracted from one recording are highly correlated; splitting them across training and test sets inflates accuracy through data leakage.
- **Objective:** classify drone presence, family, and operating condition from the DroneRF dataset under a leakage-aware evaluation.

Proposed Methodology



Three classification targets are decoded from a single 5-digit BUI filename code: presence (2 classes), family (4), and operating mode (10).

Dataset and Feature Extraction

DroneRF dataset

Class	BUI	Modes
Background	00000	1
Parrot Bebop	10000–10011	4
Parrot AR	10100–10111	4
DJI Phantom	11000	1

227 recording pairs (low `L` + high `H` sub-bands), 2.4 GHz ISM band.

Feature extraction

- Segment length $L = 100{,}000$ samples.
- 2048-point FFT magnitude, upper 1024 bins:

$$X_f = \left| \mathrm{FFT}_M(x - \bar{x}) \right|$$

- Low and high sub-bands joined and squared to a power spectrum.
- **2048-D** vector per segment; **22,700** vectors total.

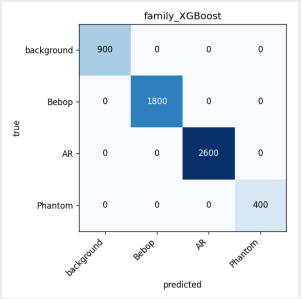
Classification Models and Rationale

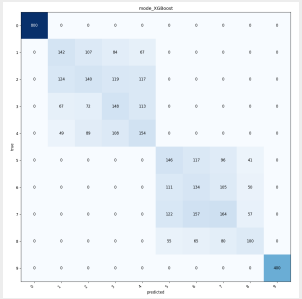
Model	**Rationale for inclusion**	**Observed behaviour**
Logistic Regression	Linear baseline; a strong result implies a separable task	Perfect on presence and family; weak on mode (0.375)
Random Forest	Bagging ensemble; captures non-linear structure	Stable; marginally lower on family (0.898)
SVM (RBF kernel)	Models curved decision boundaries	Strong on simple tasks; limited on mode (5k-sample cap)
XGBoost	Gradient-boosted trees; state-of-the-art on tabular data	**Best overall**, and decisively best on the mode task

All four models perform near-identically on the easy tasks. They separate only on the 10-class operating-mode task, where XGBoost leads, which is where the comparison is meaningful.

Classification Results and Confusion Analysis

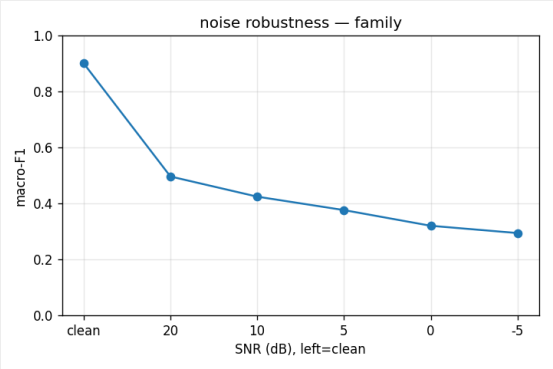
Task	LR	RF	SVM	XGBoost
Presence	1.000	0.978	1.000	1.000
Family	0.999	0.898	0.977	**1.000**
Mode (5-fold)	0.375	0.378	0.160	**0.492**





Macro-F1 by model (top). XGBoost confusion matrices: family is a clean diagonal; the 10-class mode task shows residual confusion.

Robustness Under Noise and Interference



SNR	Macro-F1
clean	0.900
20 dB	0.496
10 dB	0.424
5 dB	0.376
0 dB	0.320
−5 dB	0.294

Gaussian noise added to the family test set. Accuracy halves by 20 dB SNR, confirming that spectral features alone are not robust to the interference present in the shared band.

Advantages and Limitations

Advantages

- Passive detection without line of sight or lighting constraints.
- Interpretable, low-cost models that train quickly.
- Leakage-aware evaluation yields trustworthy metrics.
- One pipeline answers all three classification questions.

Limitations

- Operating-mode classification remains difficult (best F1 0.492).
- Spectral features degrade sharply under heavy noise.
- Only three drone families are represented in the dataset.
- Dataset-based study; no live hardware or field deployment.

Applications and Conclusion

Applications

- Passive drone surveillance near airports and restricted zones.
- Early warning for unauthorised drones at public events.
- Low-cost monitoring on software-defined radios.
- Drone-type identification for threat assessment.

Conclusion

- Four classical models were compared across three tasks under a leakage-aware split.
- Presence and family are reliably classified; XGBoost leads the hard mode task at 0.492.
- A noise test quantified feature robustness under interference.
- Classical models remain competitive with deep learning on this dataset.

Literature Survey

Paper	Method	Relevance to this work
Al-Sa'd et al., 2019 (FGCS)	Deep neural networks	Introduced the DroneRF dataset and the three-task structure adopted here.
Al-Emadi et al., 2020	XGBoost, three tasks	Closest prior work; confirms XGBoost's strength and the difficulty of the mode task.
Ezuma et al., 2020 (OJCOMS)	ML under Wi-Fi/BT interference	Motivates the noise-robustness analysis in the shared 2.4 GHz band.
Medaiyese et al., 2022 (PMC)	Wavelet features + ML	Principal alternative to the FFT/PSD features used in this project.
LUMEN, 2026 (Sensors)	PSD features, lightweight net	Validates the power-spectral feature choice on constrained hardware.
Robustness study, 2024 (Sensors)	RF UAV detector robustness	Establishes SNR-based degradation testing as standard evaluation practice.

Research gap: prior work is largely deep-learning-based or omits an explicit leakage check. This project compares four interpretable classical models across all three tasks under a recording-grouped split.

References

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